

YUCHEN YANG

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College of Information Sciences and Technology
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EDUCATION

Ph.D.	Johns Hopkins University <i>Computer Science</i> Advisor: <i>Dr. Yinzhi Cao</i>	2021 – 2025 MD, United States
M.S.	Johns Hopkins University <i>Security Informatics</i> Advisor: <i>Dr. Yinzhi Cao</i>	2019 – 2021 MD, United States
B.E.	Shandong University <i>Software Engineering</i>	2015 – 2019 Shandong, China

INTERESTS

Computer Security, Trustworthy AI

PUBLICATIONS

PEER-REVIEWED CONFERENCE PAPERS (* INDICATES CO-FIRST AUTHORS)

- 2025 **CertPHash: Towards Certified Perceptual Hashing via Robust Training**
Yuchen Yang, Qichang Liu, Christopher Brix, Huan Zhang, Yinzhi Cao
In the proceedings of USENIX Security Symposium (Usenix), 2025
- 2024 **Follow the Rules: Reasoning for Video Anomaly Detection with Large Language Models**
Yuchen Yang, Kwonjoon Lee, Behzad Dariush, Yinzhi Cao, Shao-Yuan Lo
In the Proceedings of the European Conference on Computer Vision (ECCV), 2024
- SafeGen: Mitigating Sexually Explicit Content Generation in Text-to-Image Models**
Xinfeng Li*, **Yuchen Yang***, Jiangyi Deng*, Chen Yan, Yanjiao Chen, Xiaoyu Ji, Wenyuan Xu
In the Proceedings of the ACM Conference on Computer and Communications Security (CCS), 2024
- RippleCOT: Amplifying Ripple Effect of Knowledge Editing in Language Models via Chain-of-Thought In-Context Learning**
Zihao Zhao, **Yuchen Yang**, Yijiang Li, Yinzhi Cao
In the Findings of the Association for Computational Linguistics: Empirical Methods in Natural Language Processing (EMNLP), 2024
The first author finished the paper mainly under my mentoring
- SneakyPrompt: Jailbreaking Text-to-image Generative Models**
Yuchen Yang, Bo Hui, Haolin Yuan, Neil Gong, Yinzhi Cao
In the Proceedings of the IEEE Symposium on Security and Privacy (S&P), 2024
Media Coverage (Selected): MIT Technology Review [🔗](#)
- 2023 **PrivateFL: Accurate, Differentially Private Federated Learning via Personalized Data Transformation**
Yuchen Yang*, Bo Hui*, Haolin Yuan*, Neil Gong, Yinzhi Cao
In the proceedings of USENIX Security Symposium (Usenix), 2023

Earned all Artifact Badges: Artifacts Available, Artifacts Functional, Results Reproduced

Fortifying Federated Learning against Membership Inference Attacks via Client-level Input Perturbation

Yuchen Yang, Haolin Yuan, Bo Hui, Neil Gong, Neil Fendley, Philippe Burlina, Yinzhi Cao.

In the proceedings of IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2023

2022 **Addressing Heterogeneity in Federated Learning via Distributional Transformation**

Haolin Yuan*, Bo Hui*, **Yuchen Yang***, Philippe Burlina, Neil Gong, Yinzhi Cao

In the proceedings of the European Conference on Computer Vision (ECCV), 2022

2021 **Practical Blind Membership Inference Attack via Differential Comparisons**

Bo Hui*, **Yuchen Yang***, Haolin Yuan*, Philippe Burlina, Neil Gong, Yinzhi Cao

In the proceedings of Network & Distributed System Security Symposium (NDSS), 2021

PREPRINTS

2025 **Jailbreaking Safeguarded Text-to-Image Models via Large Language Models**

Zhengyuan Jiang, Yuepeng Hu, **Yuchen Yang**, Yinzhi Cao, Neil Gong

Under review, 2025

Pseudo-Probability Unlearning: Towards Efficient and Privacy-Preserving Machine Unlearning

Zihao Zhao, Yijiang Li, **Yuchen Yang**, Wenqing Zhang, Nuno Vasconcelos, Yinzhi Cao

Under review, 2025

EXPERIENCES

The Pennsylvania State University, Assistant Professor

PA, United States

2025.07 - Current

Johns Hopkins University, Research Assistant

MD, United States

2020.03 - 2025.05

- Working on trustworthy AI/ML advised by [Dr. Yinzhi Cao](#)
- Research outcomes: publications at top-tier security conferences–S&P, Usenix, CCS, and NDSS, and vision/NLP conferences such as ECCV and EMNLP

Honda Research Institute, Research Internship

CA, United States

2023.10 - 2024.02

- Working on LLMs for video anomaly detection advised by [Dr. Shao-yuan Lo](#), [Dr. Kwonjoon Lee](#), and [Dr. Behzad Dariush](#)
- Research outcomes: a publication at ECCV 2024 and filing of a U.S. patent

Shandong Univeristy, Research Assistant

Shandong, China

2018.12 - 2019.06

- Working on an NLP-driven auto-grading system for Chinese composition in primary and secondary education advised by [Dr. Yuqing Sun](#)
- Research outcomes: my undergraduate dissertation: Automatic Grading System for Chinese Composition via Bi-directional LSTM

Chinese Academy of Sciences, Research Internship

Beijing, China

2018.06 - 2018.09

- Working on developing SVM+ by distinguishing the privileged vectors within SVM (support vector machine) algorithm advised by [Dr. Yingjie Tian](#)
- Research outcomes: a patent filed under IP Australia

AWARDS AND HONORS

Selected as ICLR Notable Reviewer

2025.05

Selected to attend Individualized Cybersecurity Research Mentoring Workshop (iMentor)

2021.11

Academic Scholarship, Shandong University

2018.09

MEDIA COVERAGE

Text-to-image AI models can be tricked into generating disturbing images, MIT Technology Review 	2023.11
AI art generators can be fooled into making NSFW images, IEEE Spectrum 	2023.11
Nonsense prompts trick AIs into producing NSFW images, Technology Networks 	2023.11
Researchers reveal vulnerabilities in AI models, prompting concerns, Cryptopolitan 	2023.11
AI generators can be tricked into making NSFW content, JHU Engineering School News 	2023.11
SneakyPrompt: Revealing the vulnerabilities of text-to-image AI, The Johns Hopkins News-Letter 	2023.12

PROFESSIONAL SERVICES

CONFERENCE/JOURNAL REVIEWING

PC , IEEE Symposium on Security and Privacy (S&P)	2026
ACM Conference on Computer and Communications Security (CCS)	2025
ACM Workshop on Adaptive and Autonomous Cyber Defense (AACD)	2024
AEC , IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)	2024
Reviewer , International Conference on Learning Representations (ICLR)	2025
International Conference on Computer Vision (ICCV)	2025
IEEE Transactions on Dependable and Secure Computing (TDSC)	2023, 2024
IEEE Transactions on Information Forensics & Security (T-IFS)	2024
External Reviewer , IEEE Symposium on Security and Privacy (S&P)	2025
USENIX Security Symposium	2023, 2024
ACM Conference on Computer and Communications Security (CCS)	2022
ACM ASIA Conference on Computer and Communications Security (ASIACCS)	2024
IEEE Computer Security Foundations Symposium (CSF)	2022, 2024
IEEE International Conference on Distributed Computing Systems (ICDCS)	2022

ORGANIZING AND CHAIRING

Session Chair , IEEE Workshop on Deep Learning Security and Privacy (DLSP)	2024
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TEACHING AND ADVISING

TEACHING

Teaching assistant, Web Security	Fall 2020, 2022
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ADVISING

Zihao Zhao, undergraduate student at JHU	2023.12 - 2024.09
Qichang Liu, undergraduate student at Tsinghua University	2024.06 - 2024.09
Yichen Li, undergraduate student at Beijing University	2024.06 - 2024.09
Adila Abudurehman, master student at JHU. Now: Software engineer at Microsoft	2023.10 - 2024.02
Manoj Valeti, master student at JHU. Now: Software engineer at AWS Amazon	2023.10 - 2024.02
Amodini Vardhan, master student at JHU. Now: Security engineer at Twilio	2023.10 - 2024.02
Anning Li, undergraduate student at UESTC. Now: Master student at CMU	2023.06 - 2023.09
Ashi Garg, master student at JHU. Now: Research associate at HLTCOE	2022.09 - 2023.03

PATENT

2024	System and Method Using Reasoning for Video Anomaly Detection with Large Language Models Yuchen Yang , Kwonjoon Lee, Shao-Yuan Lo, Behzad Dariush <i>Under US Patent Application (Filed), 2024</i>
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2018 **A New System for Stock Volatility Prediction by Using Privileged Support Vector Machines**
Yuchen Yang, Zheng Yan, Haoyang Li, Zhixuan Lv, Weiwei Zhao, Simiao Zhao
Under IP Australia Application (No.2018101304), 2018

INVITED TALKS

"Diagnose, Correct, Steer: Towards Functional and Trustworthy AI Systems", The Pennsylvania State University, Invited talk	2025.3
"Diagnose, Correct, Steer: Towards Functional and Trustworthy AI Systems", Case Western Reserve University, Invited talk	2025.2
"Diagnose, Correct, Steer: Towards Functional and Trustworthy AI Systems", Rice University, Invited talk	2025.2
"Diagnose, Correct, Steer: Towards Functional, Trustworthy AI", Virginia Tech, Invited talk	2025.2
"Diagnose, Correct, Steer: Towards Functional, Trustworthy AI", Rochester Institute of Technology, Invited talk	2025.1
"Diagnose, Correct, Steer: Towards Functional, Trustworthy AI", Monash University, Invited talk	2024.10
"Zero-shot Video Anomaly Detection: Steering LLMs for Rules-based Reasoning", Voxel51, Invited talk	2024.11